AxisField-GS: Feed-Forward Articulation Recovery from Single-State Semantic Gaussian Assets

Anonymous ACCV 2026 Submission

Paper ID #290

Abstract. Static 3D Gaussian assets provide compact and photorealistic digital replicas, but they do not encode the kinematic structure required for interaction or animation. We study a lightweight digital-asset upgrade problem: recovering the primary revolute joint of a pre-trained single-state semantic 3D Gaussian asset and converting the static representation into an articulated Gaussian asset. The proposed AxisField-GS predicts a dense articulated field over Gaussians, including movable confidence, support confidence, per-Gaussian axis offsets, and local axis directions. Instead of regressing a joint axis from a single global feature, AxisField-GS aggregates local predictions through self-consistent axis voting, where movable confidence, offset magnitude, and direction consistency jointly determine which Gaussians participate in line fitting. This design aligns the training objective with the inference-time estimator and prevents the final axis from relying on movable segmentation alone. On PartNet-Mobility-derived semantic Gaussian assets, AxisField-GS achieves 0.063 mean normalized symmetric axis distance, 2.43 degrees mean angular error, and 0.864 movable IoU on the test split, while requiring only one feed-forward pass over a single static asset. Ablations show that removing movable confidence, offset proximity, or direction consistency substantially degrades axis recovery, supporting the need for self-consistent dense geometric voting.

Keywords: 3D Gaussian Splatting · Articulated Object Modeling · Articulated Digital Assets · Feed-Forward Articulation Recovery

1 Introduction

Recent radiance-field and explicit neural scene representations have made photorealistic digital asset construction increasingly practical [1, 3, 4, 11, 21, 24, 27, 37]. Semantic assets further attach CLIP, DINO/DINOv2, SAM, and language features to radiance or Gaussian fields for querying, segmentation, editing, and mesh extraction [2, 7, 12, 14, 28, 35, 36, 46]. Yet a visually faithful Gaussian asset is not necessarily actionable. Many everyday objects, such as cabinets, laptops, washing machines, and containers, are defined not only by appearance but also by kinematic structure: a static Gaussian asset can render the observed state, but it does not reveal which part can move, where the revolute axis lies, or how to synthesize a new articulated state.

This distinction matters for asset libraries and embodied perception systems, where a static object asset may already exist before editing, simulation, animation, or robot interaction is considered. A useful upgrade procedure should operate directly on the existing Gaussian set, avoid new multi-state captures or manual rigging, and return articulation in a form that can be applied back to the same representation.

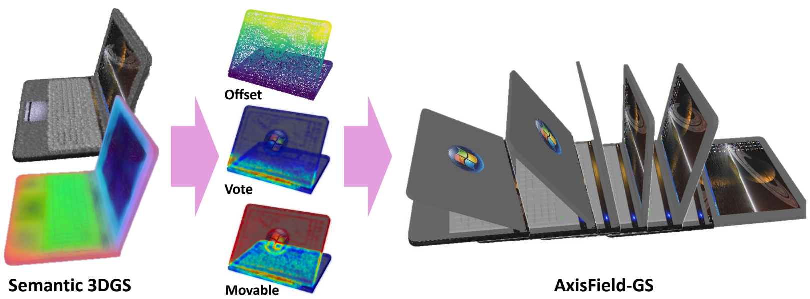


Fig. 1: AxisField-GS Given a single-state semantic 3DGS, AxisField-GS predicts dense articulated fields over Gaussians, visualized here by axis offsets, self-consistent vote weights, and movable confidence. These dense cues are aggregated to recover a revolute axis directly in Gaussian space, upgrading a static visual asset into an articulated digital asset for hinge-motion synthesis without observed motion, multi-state input, or per-object optimization.

Existing articulated-object and digital-twin methods infer articulation from images, depth, point clouds, interactions, meshes, or multiple motion states [5, 10, 15–17, 20, 22, 25, 38, 40, 41], and recent Gaussian methods often reconstruct articulated assets from multi-state or multi-view observations with per-object optimization [8, 13, 18, 23, 42, 47]. These settings are powerful when motion evidence is available, but they do not isolate the single-state asset-upgrade setting: a semantic Gaussian asset has already been reconstructed, and the downstream system must infer how to move it without collecting additional articulated states.

We therefore study a narrower digital-asset upgrade problem: given a single-state semantic 3D Gaussian asset, recover one dominant revolute joint and its Gaussian-level movable field. We do not claim general multi-joint recovery, arbitrary kinematic-tree reconstruction, or complete physical digital-twin modeling. The challenge is that articulation is not observed as motion; the model must infer the joint from geometric layout, part appearance, semantic features, and local support structure on an unordered set of Gaussians.

Our key observation is that the revolute axis should not be treated as a single global regression target. Even from a static asset, local Gaussians provide weak but redundant evidence about the movable part and its relation to the support. AxisField-GS therefore predicts a dense articulated field: each queried Gaussian estimates movable confidence, support confidence, a local offset toward the axis, and a local axis direction. The final axis is recovered by self-consistent voting and weighted line fitting, where a Gaussian receives high weight only when it is likely movable, predicts a plausible near-axis offset, and agrees with the dominant direction hypothesis.

This paper makes three contributions. First, it formulates feed-forward primary articulation recovery as a lightweight digital-asset upgrade problem that converts a pre-trained single-state semantic 3D Gaussian asset into a Gaussian-native articulated asset. Second, it introduces a dense articulated field architecture that combines a Point Transformer backbone, joint query decoding, and full-Gaussian point decoding to predict movable, support, offset, and direction fields. Third, it introduces self-consistent axis voting, an inference rule and training objective that encourage the dense predictions to agree on a single revolute line. Experiments on PartNet-Mobility-derived semantic Gaussian assets quantify this single-state feed-forward setting, while ablations demonstrate that axis recovery requires the joint use of movable confidence, offset geometry, and direction consistency.

2 Related Work

Articulated digital twins and articulation recovery. Prior work estimates articulated parts, part poses, actionable interaction cues, or digital twins from shapes, RGB-D observations, point clouds, interaction pairs, monocular images, or static meshes [5,6,10,15–17,20,22,25,38,40,41]. Generative methods such as SINGAPO further study articulated part generation from a single image [19]. These works show that articulation can be inferred from visual and geometric evidence, but their inputs are raw observations, generated parts, meshes, or interaction evidence rather than an existing semantic Gaussian asset.

Articulated Gaussian assets. Gaussian-based articulated reconstruction methods are the closest family in representation space. Deformable radiance fields recover dynamic geometry from time-varying observations [29–31], while ScrewSplat, ArtGS, ArticulatedGS, REArtGS, SplArt, Part2GS, SPAGS, and related methods reconstruct or optimize articulated Gaussian assets from multi-view, multi-state, sparse-view, or video observations [8, 13, 18, 23, 42, 43, 47]. These works usually couple articulation discovery with reconstruction or per-object optimization. In contrast, AxisField-GS assumes that a semantic 3D Gaussian asset has already been reconstructed, and predicts articulation feed-forward from one static state.

Thus, the closest prior methods differ in test-time evidence and optimization. Multi-state Gaussian methods observe motion or multiple configurations; per-object methods adapt to each test instance. AxisField-GS instead receives one static semantic Gaussian asset and reports Gaussian-space baselines as contextual, not identical-information, comparisons.

Dense voting and point-set reasoning. Our method is also related to point-set architectures and dense voting. PointNet, PointNet++, DGCNN, Point Transformer, and Point Transformer V3 provide backbones for unordered 3D data [33, 34, 39, 44, 48]; VoteNet and PointGroup use point-wise votes or offsets for object localization and grouping [9, 32]. AxisField-GS applies this dense-estimation philosophy to an unoriented 3D line representing the primary revolute axis, where useful votes must be movable, near-axis, and directionally consistent.

3 Method

3.1 Problem Formulation

A semantic Gaussian asset is a static 3D Gaussian Splatting asset whose Gaussians carry both geometric attributes and a learned semantic feature. No motion pair, multi-state observation, mesh topology, or test-time interaction is provided. Let such an asset be denoted by

$$\mathcal{G} = \{g_i\}_{i=1}^{N}, \quad g_i = (\mu_i, q_i, s_i, \alpha_i, n_i, f_i), \tag{1}$$

where $\mu_i \in \mathbb{R}^3$ is the Gaussian center, $q_i \in \mathbb{S}^3$ is its unit-quaternion rotation, $s_i \in \mathbb{R}^3$ is the anisotropic scale, α_i is opacity, $n_i \in \mathbb{R}^3$ is an estimated normal, and $f_i \in \mathbb{R}^{32}$ is a semantic feature obtained by projecting DINOv2 features to 32 dimensions [28]. The asset is normalized into an object coordinate frame before being processed by the network. The goal is to recover the primary revolute joint

$$\mathcal{A} = (o, v), \quad o \in \mathbb{R}^3, \quad v \in \mathbb{S}^2, \tag{2}$$

together with a Gaussian-level movable field. The line direction is unoriented, so v and $-v$ represent the same axis. We focus on one dominant revolute joint per input asset and do not claim multi-joint set prediction or full kinematic-tree reconstruction.

3.2 Network Overview

AxisField-GS processes a semantic Gaussian asset in three stages, illustrated in Fig. 2. First, a memory-efficient shortlist of 8192 Gaussians is sampled from the asset and encoded by a Point Transformer V3 backbone [44]. The backbone produces multi-scale point memories that preserve local geometric cues while providing object-level context. Second, a joint query decoder uses a small set of learnable joint queries to aggregate global articulation evidence. Although only one primary joint is supervised in this paper, the query structure provides a structured latent space for joint-level reasoning and supplies auxiliary joint-level axis predictions during training. Third, a full-Gaussian point decoder reads the multi-scale memories and joint context to predict dense fields for query Gaussians. During training, the decoder is supervised on 2048 sampled query Gaussians; during inference, it is applied to the full Gaussian asset in chunks of

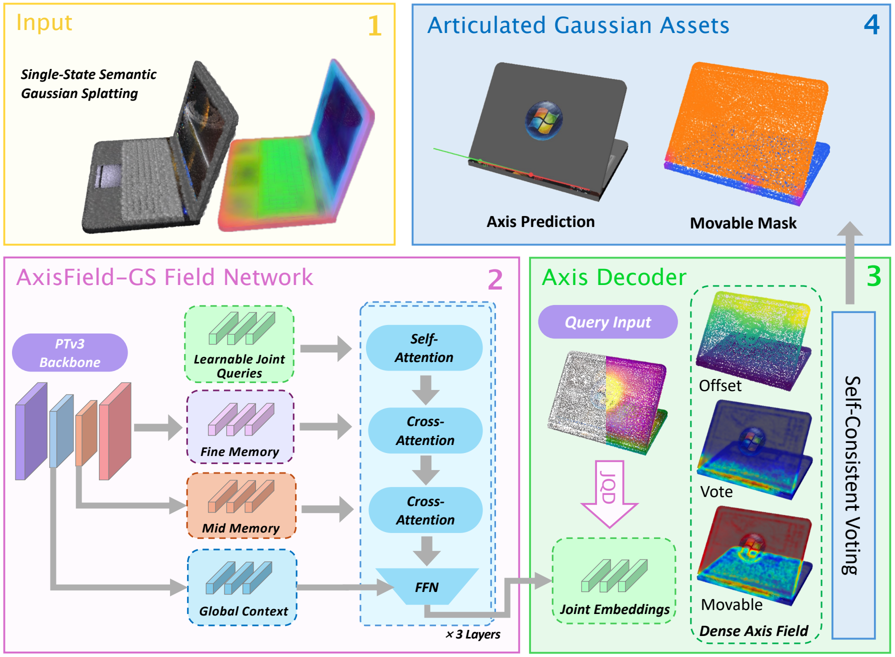


Fig. 2: AxisField-GS overview. Given a single-state semantic 3D Gaussian asset, AxisField-GS predicts dense movable, offset, and vote fields with a PTv3 backbone and joint-query decoder. Self-consistent voting then selects reliable Gaussian evidence to fit the axis, producing an articulated asset with a recovered axis and movable mask.

4096 Gaussians. Joint-level predictions provide training and contextual signals, while the reported test-time axis is recovered from dense self-consistent voting.

The joint query decoder is a three-layer Transformer-style decoder over $K = 4$ learnable joint queries, rather than a single cross-attention layer. Each layer first performs self-attention among the joint queries, then lets the queries cross-attend to fine and mid-level PTv3 memories, and finally updates them with a global-conditioned feed-forward block. The model also refines the top-level PTv3 memory with two global self-attention layers before forming the global context; this global memory refinement is separate from the joint query decoder itself.

The point decoder is implemented as a per-Gaussian memory readout. Each query Gaussian is encoded from its geometric attributes and semantic feature, retrieves local fine and mid-level memory neighborhoods by nearest-neighbor lookup in normalized object coordinates, and uses cross-attention to read those memory tokens. The resulting point token then attends to the joint-query embeddings and to the refined global memory before a global-conditioned MLP produces the final hidden state. The movable, support, offset, and direction predictions are produced by small MLP heads, with the direction head normalized to the unit sphere.

3.3 Dense Articulated Field

For each query Gaussian at normalized position p_i, the network predicts four dense quantities:

$$\hat{m}_i, \quad \hat{b}_i, \quad \hat{\Delta}_i \in \mathbb{R}^3, \quad \hat{d}_i \in \mathbb{S}^2. \tag{3}$$

Here $\hat{m}_i$ is the movable logit, $\hat{b}_i$ is an auxiliary support logit, $\hat{\Delta}_i$ is an offset from the Gaussian center toward the joint axis, and $\hat{d}_i$ is a local axis-direction prediction. The predicted axis anchor for point i is

$$\hat{a}_i = p_i + \hat{\Delta}_i. \tag{4}$$

The movable and support fields provide local semantic and structural cues, while offsets and directions provide geometric evidence for the final axis. The support target is generated from movable Gaussians near the ground-truth axis using a Gaussian band in normalized object coordinates. It is used only as auxiliary dense supervision and is not used as an oracle vote mask at inference; the final inference rule is based on self-consistent predicted weights rather than support labels or a single joint-token regression. This distinction is essential for the single-state setting, because a high-quality movable mask is not by itself an axis estimator. Large movable surfaces can dominate a naive fit even when they are far from the hinge, so the model must also learn which movable Gaussians provide geometrically useful axis evidence.

3.4 Self-Consistent Axis Voting

The central design choice in AxisField-GS is to recover the axis from dense, mutually consistent predictions. A point should contribute to line fitting only when it is likely movable, predicts a compact offset to the axis, and agrees with the dominant direction. This rule is intentionally more restrictive than thresholding the movable field: it suppresses static supports, distant movable surfaces, and directionally inconsistent anchors before the final line fit. We compute a self-consistent vote weight

$$\hat{w}_i = \sigma(\hat{m}_i)\, \phi_{\mathrm{off}}(\hat{\Delta}_i)\, \phi_{\mathrm{dir}}(\hat{d}_i), \tag{5}$$

where $\sigma(\hat{m}_i)$ is the movable probability. The offset term downweights anchors predicted from points far from the axis,

$$\phi_{\mathrm{off}}(\hat{\Delta}_i) = \exp\left(-\frac{\|\hat{\Delta}_i\|_2^2}{2\sigma_o^2}\right), \tag{6}$$

with $\sigma_o = 0.08$ in our experiments. Direction consistency is computed without choosing an arbitrary sign. We first estimate a dominant unoriented direction using preliminary weights $\tilde{w}_i = \sigma(\hat{m}_i)\phi_{\mathrm{off}}(\hat{\Delta}_i)$:

$$\hat{u} = \mathrm{eig}_{\max}\left(\sum_i \tilde{w}_i\, \bar{d}_i \bar{d}_i^\top\right), \quad \bar{d}_i = \frac{\hat{d}_i}{\|\hat{d}_i\|_2 + \epsilon}. \tag{7}$$

We then set $\phi_{\mathrm{dir}}(\hat{d}_i) = |\hat{d}_i^\top \hat{u}|$. The same self-consistent weighting rule is used during training and inference, which reduces the train-test mismatch caused by training with oracle vote weights and inferring with predicted weights. Inference fits the line from the highest-confidence votes, keeping the top 256 votes and any additional votes with weight at least 0.05; if no vote passes the threshold, the top-weighted votes provide the fallback.

Given anchors $\{\hat{a}_i\}$ and weights $\{\hat{w}_i\}$, the axis origin is represented by the weighted centroid

$$\hat{c} = \frac{\sum_i \hat{w}_i \hat{a}_i}{\sum_i \hat{w}_i + \epsilon}. \tag{8}$$

The axis direction is recovered by the top eigenvector of the weighted covariance matrix

$$\hat{\Sigma} = \frac{\sum_i \hat{w}_i (\hat{a}_i - \hat{c})(\hat{a}_i - \hat{c})^\top}{\sum_i \hat{w}_i + \epsilon}, \quad \hat{v} = \mathrm{eig}_{\max}(\hat{\Sigma}). \tag{9}$$

This weighted line fitting is differentiable during training through the predicted anchors and weights, and it is used directly as the final inference mechanism. The same dense-voting rule is used for all quantitative and qualitative results. Because the line is estimated from anchors rather than from a pre-defined hinge part, the method can combine evidence from spatially distributed Gaussians while remaining invariant to the arbitrary ordering of the asset.

3.5 Training Objective

The full training objective combines dense per-Gaussian supervision with axis-level supervision:

$$\mathcal{L} = \mathcal{L}_{\mathrm{mov}} + \lambda_b \mathcal{L}_{\mathrm{sup}} + \lambda_\Delta \mathcal{L}_{\mathrm{off}} + \lambda_d \mathcal{L}_{\mathrm{dir}} + \lambda_a \mathcal{L}_{\mathrm{axis}}. \tag{10}$$

The movable and support terms use focal binary cross-entropy to handle class imbalance. The offset loss supervises the nearest-point vector from a query Gaussian to the ground-truth axis, and the direction loss uses the unoriented cosine distance $1 - |\hat{d}_i^\top v^\star|$. The axis-level loss is applied after a warmup period so that dense predictions become stable before differentiable line fitting is emphasized:

$$\mathcal{L}_{\mathrm{axis}} = d_{\mathrm{sym}}((\hat{c}, \hat{v}), (o^\star, v^\star)) + \lambda_\theta \left(1 - |\hat{v}^\top v^\star|\right). \tag{11}$$

Here d_{sym} is the symmetric point-to-line distance in normalized object coordinates. This loss directly supervises the axis recovered by predicted self-consistent weights, rather than only supervising per-point offsets under oracle weighting. The axis loss is activated after 60 warmup epochs so that dense predictions are stable before the differentiable line-fitting objective is emphasized. The objective therefore connects local supervision and global articulation quality: movable and support losses teach the model where evidence may come from, offset and direction losses teach what each local vote should predict, and the axis loss penalizes the final line produced by the same estimator used at test time.

4 Experiments

4.1 Experimental Setup

We evaluate on semantic 3D Gaussian assets trained from PartNet-Mobility-derived articulated objects [26, 45]. Each asset represents a single object state and stores Gaussian centers, unit-quaternion rotations, anisotropic scales, opacities, estimated normals, and a 32-dimensional DINOv2 semantic feature after PCA compression. The dataset covers nine object categories: StorageFurniture, Laptop, TrashCan, Dishwasher, Safe, Display, WashingMachine, Microwave, and Phone. The train, validation, and test splits contain 1992, 256, and 632 single-state assets, respectively, with validation and test containing 32 and 79 base objects. Splits are constructed at the base-object level so that validation and test assets come from objects unseen during training. The present paper focuses on revolute joints and primary-joint recovery; when an object contains multiple joints, each evaluated sample specifies the primary target joint and its corresponding movable field.

This evaluation protocol is designed to test category-level generalization rather than memorization of object instances. The model sees only a static asset at test time, and the target motion state is not provided as an input observation. All axis distances are computed in the normalized object coordinate frame used by the asset, and movable IoU is evaluated over valid Gaussians rather than over a reconstructed mesh. These choices keep the evaluation aligned with the intended use case: upgrading a Gaussian asset in its native representation.

The checkpoint and inference rule are selected using validation symmetric axis distance, and test results are reported once under the validation-selected setting. Unless otherwise stated, AxisField-GS uses dense self-consistent voting at inference. For contextual Gaussian-space baselines, we include ScrewSplat and SplArt runs under their native multi-state per-object optimization regimes. These baselines use different test-time evidence from AxisField-GS, which consumes one single-state semantic Gaussian asset and performs one feed-forward prediction.

Metrics. We report normalized symmetric axis distance, axis angular error, movable IoU, and success rates under axis-distance and angular thresholds. For a predicted line $(\hat{c}, \hat{v})$ and ground-truth line $(o^\star, v^\star)$ in normalized object coordinates, the symmetric distance is

$$d_{\mathrm{sym}} = \frac{1}{2}\left[d_{\mathrm{line}}(\hat{c}; o^\star, v^\star) + d_{\mathrm{line}}(o^\star; \hat{c}, \hat{v})\right], \tag{12}$$

where $d_{\mathrm{line}}(p; o, v) = \|(p-o) - ((p-o)^\top v)v\|_2$ for unit direction v. Angular error is sign-invariant and is computed as $\arccos(|\hat{v}^\top v^\star|)$ in degrees. Movable IoU is computed over valid Gaussians using a movable-probability threshold of 0.5. We report success at $d_{\mathrm{sym}} < 0.10$ and angular error below 10°, and compute confidence intervals by cluster bootstrap over base objects where intervals are reported.

4.2 Main Results

Tab. 1 reports validation and test performance. On the test split, AxisField-GS achieves a mean normalized symmetric axis distance of 0.0631, a median distance of 0.0331, a mean angular error of 2.43 degrees, and a movable IoU of 0.864. The model recovers axes within 0.10 normalized symmetric distance for 86.4% of test samples and within 10 degrees for 96.7% of test samples. The validation and test numbers are close despite the base-object split, suggesting that the model is not primarily memorizing object instances from training.

The gap between mean and median axis distance is also informative. Most predictions are tightly localized around the ground-truth axis, as reflected by the low median distance and near-zero median angular error. The higher mean distance indicates a tail of difficult objects in which the axis is plausible but spatially shifted. This pattern matches the single-state ambiguity of the task: without observed motion, the direction of a hinge can often be inferred from elongated geometry, but the exact support side or pivot location can remain uncertain for broad doors and visually symmetric parts.

Table 1: AxisField-GS performance using the validation-selected checkpoint and self-consistent dense voting. Distance, angle, and IoU columns report mean / median; success columns report percentages.

Split	n	Sym. dist. ↓	Angle ↓	IoU ↑	Sym. < 0.10 (%)	Angle < 10° (%)
Val	256	0.0667 / 0.0333	2.87 / 0.09	0.894 / 0.940	83.6	93.4
Test	632	0.0631 / 0.0331	2.43 / 0.08	0.864 / 0.932	86.4	96.7

4.3 Qualitative Object Visualizations

Fig. 3 shows representative object-level visualizations selected from the test set. Each example shows the recovered revolute axis from multiple viewpoints and the dense fields that support the prediction, including the support response, joint-query attention, offset field, and movable confidence. These visualizations complement the aggregate metrics by showing how the method localizes the hinge axis from spatially distributed Gaussian evidence rather than from a single global regression output. In successful cases, high-confidence movable Gaussians cover the articulated part, but the strongest axis evidence is concentrated near geometrically consistent regions instead of being spread uniformly over the whole movable surface. This behavior is the intended effect of combining movable confidence, offset proximity, and direction consistency in the final vote.

4.4 Comparison Across Input and Optimization Regimes

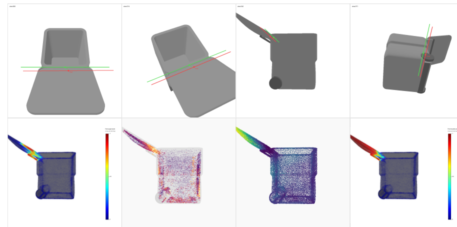
Tab. 2 contextualizes AxisField-GS against Gaussian-space articulated reconstruction methods under different input and optimization regimes. AxisField-GS consumes one single-state semantic Gaussian asset and predicts articulation in a feed-forward pass. ScrewSplat and SplArt instead optimize each object from multiple observed states [13, 18]. These rows are therefore not identical-information comparisons; they make the evaluation regime explicit and locate our task relative to existing Gaussian-space approaches. The result shows that a category-level feed-forward model, when trained for the single-state asset-upgrade problem, can produce accurate Gaussian-space axes without test-time motion evidence or per-object optimization.

The baseline numbers should be interpreted through their native assumptions. Multi-state methods jointly optimize reconstruction and motion, whereas AxisField-GS is trained and evaluated directly on normalized semantic Gaussian assets. Thus, the table is a contextual comparison across input and optimization regimes rather than a controlled replacement of one module within the same pipeline. Nevertheless, the comparison is informative: even with observed motion and per-object optimization, the contextual baselines yield larger axis-distance and angular errors on this benchmark, while AxisField-GS remains accurate from a single static asset. This suggests that directly training the dense field and final voting rule for the asset-upgrade interface is important for producing axes that match the reported normalized line metrics. The comparison therefore does not claim that the baselines are weaker methods in their intended settings; rather, it shows that the asset-upgrade setting requires a different inference interface.

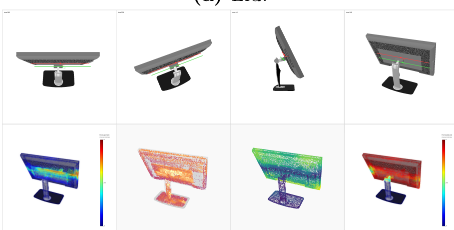
Table 2: Gaussian-space contextual comparison on the test set. The table reports mean metrics under each method's native input and optimization regime; rows are not identical-information comparisons.

Method	Evidence	Test-time regime	Motion	Sym. ↓	Angle ↓	IoU ↑
AxisField-GS	1 semantic state	feed-forward	no	0.063	2.43	0.864
ScrewSplat [13]	2 states	per-object opt.	yes	0.422	28.86	0.088
ScrewSplat [13]	3 states	per-object opt.	yes	0.301	21.81	0.134
ScrewSplat [13]	5 states	per-object opt.	yes	0.280	19.68	0.164
SplArt [18]	2 states	per-object opt.	yes	0.569	39.03	0.276

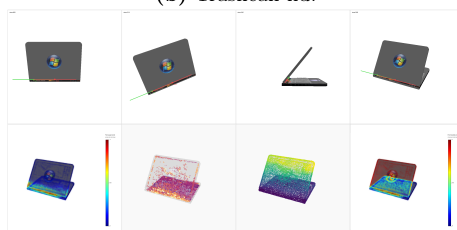
(a) Lid.



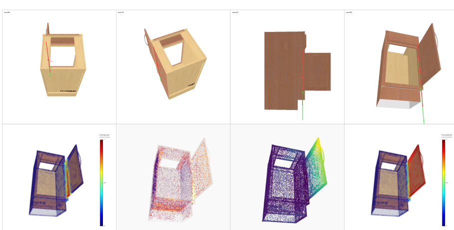
(b) Trashcan lid.



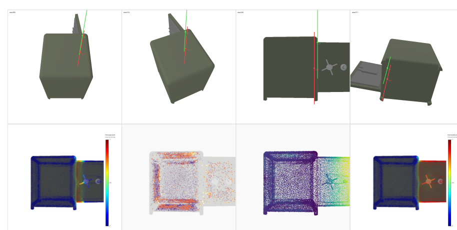
(c) Display screen.



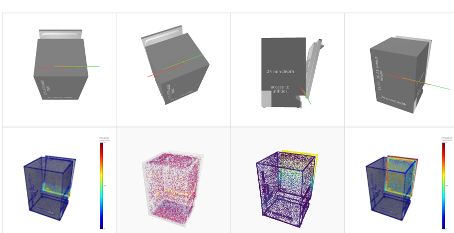
(d) Side screen.



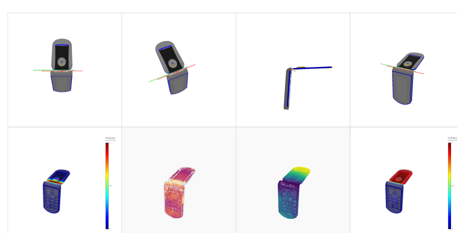
(e) Cabinet door.



(f) Safe door.



(g) Dishwasher door .

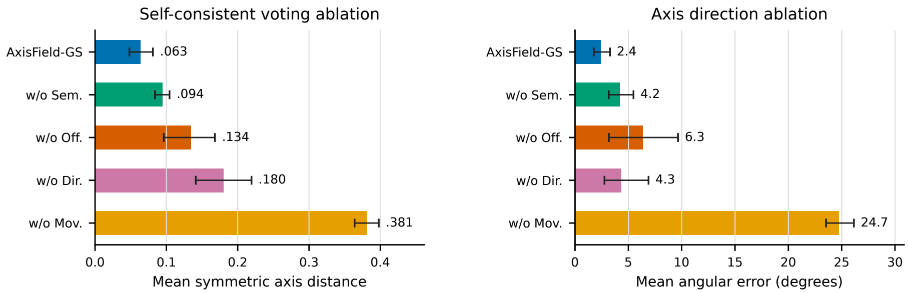


(h) Flip phone.

Fig. 3: Representative object-level visualizations. In each subfigure, the top row overlays predicted and ground-truth revolute axes from multiple viewpoints, while the bottom row shows support, attention, offset, and movable fields used by AxisField-GS to assemble the final dense-voting estimate.

4.5 Ablation Studies

The ablations isolate the two design decisions that are specific to AxisField-GS. The first is the use of semantic Gaussian features: a geometry-only model removes the DINOv2-derived feature channel during both training and inference, testing whether Gaussian positions, scales, opacity, normals, and color-like attributes are sufficient for single-state hinge reasoning. The second is the self-consistent vote weight in Eq. (5): starting from the same validation-selected AxisField-GS checkpoint, we disable one multiplicative factor at a time during line fitting while leaving the learned dense fields unchanged. This protocol separates representation learning from inference-time vote aggregation, so the vote-factor rows measure whether the final axis requires joint agreement among movable confidence, offset proximity, and direction consistency. It also avoids attributing a vote-aggregation failure to retraining noise, because the dense predictions are identical for the three vote-factor interventions.



(a) Mean symmetric axis distance.

(b) Mean angular error.

Fig. 4: Voting and semantic ablation plots on the test split. Short labels match Tab. 3; error bars denote the available confidence intervals.

Table 3 and Fig. 4 show that semantic features mainly improve localization and movable-field quality. Removing semantics increases mean symmetric distance from 0.063 to 0.094 and reduces movable IoU from 0.864 to 0.770. However, the angular success rate remains high at 95.7%, indicating that Gaussian geometry alone often preserves enough coarse directional evidence. The value of the semantic channel is therefore not simply that it points the axis in the right direction; rather, it helps identify which part-level evidence should participate in the final line fit.

The vote-factor ablations explain why the final estimator uses a product of three cues instead of a single dense prediction. Removing movable confidence is the most damaging intervention: mean symmetric distance rises to 0.381, angular error rises to 24.72 degrees, and success under $d_{\text{sym}} < 0.10$ drops from 86.4% to 15.1%. The movable IoU remains nearly unchanged in this row because the segmentation head is not modified. The failure is therefore not that the model loses its movable prediction, but that static and irrelevant Gaussians are allowed to vote during line fitting. Offset proximity and direction consistency provide complementary geometric filters. Without offset proximity, anchors far from the joint axis can dominate the fitted line, increasing symmetric distance to 0.134 and angular error to 6.35 degrees. Without direction consistency, localization degrades further to 0.180 mean symmetric distance and angular-threshold success falls to 88.4%, even when the predicted movable field remains strong. Together, these results support the central inference design: reliable single-state axis recovery requires votes that are assigned to the movable part, close to the predicted axis, and consistent with a common unoriented direction. Additional training-objective diagnostics are provided in the supplementary material.

Table 3: Ablation on the test split. w/o Sem. is trained and evaluated without semantic features. w/o Mov., w/o Off., and w/o Dir. use the same AxisField-GS checkpoint and disable one inference-time vote factor from Eq. (5). Success columns report percentages.

Variant	Sym. ↓	Angle ↓	IoU ↑	Sym. < 0.10	Angle $< 10^\circ$
AxisField-GS	0.063	2.43	0.864	86.4	96.7
w/o Sem.	0.094	4.18	0.770	63.8	95.7
w/o Off.	0.134	6.35	0.865	77.8	95.9
w/o Dir.	0.180	4.33	0.865	74.9	88.4
w/o Mov.	0.381	24.72	0.864	15.1	11.1

4.6 Category-Level Behavior

Tab. 4 summarizes part/joint-level categories on the test split. The rows cover all 632 test samples. Cabinet-door, lid, and flipping-lid samples obtain lower symmetric distances than generic door samples, which remain the most challenging group. The easier categories tend to have compact or visually well-aligned hinge structures, so the movable field and offset votes reinforce each other. Generic doors are more heterogeneous: the movable surface can be broad, the hinge side can be weakly expressed in the Gaussian geometry, and visual semantics alone may not disambiguate the correct support side. These errors are consistent with the single-state setting, where the model must infer articulation from static evidence rather than from observed displacement.

Table 4: Part/joint-level category breakdown on the test split.

Category	n	Sym. mean ↓	Angle mean ↓	IoU ↑
cabinet_door	240	0.0366	1.44	0.898
screen	40	0.0399	0.07	0.819
shaft	16	0.0506	0.09	0.779
flipping_lid	8	0.0538	0.25	0.936
lid	88	0.0557	0.54	0.911
screen_side	32	0.0584	0.09	0.823
display_screen	32	0.0780	0.13	0.985
door	176	0.1080	6.40	0.795

5 Discussion and Limitations

The experiments support three conclusions. First, dense local predictions can recover accurate primary revolute axes from many single-state Gaussian assets. Second, reliable line fitting requires the joint use of movable confidence, offset proximity, and direction consistency, since each vote factor filters a different failure mode. Third, semantic features improve the selection of useful local evidence, while geometry still provides coarse directional cues.

The main failures are wrong-side hinges on broad doors, occasional orthogonal direction estimates, and movable-field errors that downweight correct local evidence. In our test-set audit, 52 samples triggered at least one localization, angular, IoU, or confidence failure condition, with movable-field errors and confidence mismatch as the most common contributors. These cases reflect the ambiguity of recovering articulation without observed motion, especially when the shape lacks an explicit hinge, gap, handle, or support cue.

AxisField-GS should therefore be viewed as a Gaussian-space asset-upgrade module rather than a complete articulated digital-twin system. It recovers one dominant revolute joint and a movable field, but does not infer prismatic joints, multi-joint kinematic trees, joint limits, contact constraints, or physical dynamics. Extending the formulation to multiple joints will require assigning Gaussians to competing motion hypotheses and enforcing kinematic consistency.

6 Conclusion

We presented AxisField-GS, a feed-forward method that upgrades a single-state semantic 3D Gaussian asset into an articulated asset by recovering its primary revolute joint. Dense articulated fields and self-consistent voting produce accurate axes and movable fields on PartNet-Mobility-derived semantic Gaussian assets, while ablations show that movable confidence, offset proximity, and direction consistency are all necessary for reliable line fitting. The formulation is limited to one dominant revolute joint, but provides a practical interface for making static Gaussian assets movable without additional articulated states.

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